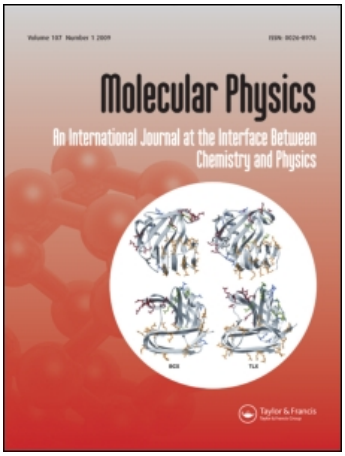
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Vibration-rotation coordinates and kinetic energy operators for polyatomic molecules

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Vibration–rotation coordinates and kinetic energy operators for polyatomic molecules

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In molecular vibration–rotation calculations it is frequently desirable to work in a nuclear coordinate system tailored to the nuclear potential energy surface. It is now possible to derive the kinetic energy operator for virtually any coordinate system by using computer algebra programs. We discuss how to choose coordinates for use in practical calculations, in particular variational energy level calculations, with special consideration of the vibration–rotation separation. As an example we present a kinetic energy operator suitable for calculating the rovibrational spectrum of sequentially bonded four-atomic molecules, using valence coordinates. We carry out a rigorous symmetry classification using the MS group, and also consider how the problem of singularities affects the choice of basis set.

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1. Introduction

There is a range of perturbative, variational, and other approximate methods that have been useful in predicting the rovibrational spectra of polyatomic molecules. These are all based on finding good approximate eigensolutions of a Hamiltonian $\hat{H}_{VR} = \hat{T}_{VR} + \hat{V}_N$ that has been written down in an appropriate set of $3N - 3$ coordinates, where N is the number of nuclei, and three translational coordinates have been separated exactly; $\hat{V}_N$ is the nuclear potential function and $\hat{T}_{VR}$ the vibration–rotation kinetic energy operator. We shall assume throughout that there are no external fields. It is usual in calculations of rovibrational states to use an approximate expansion for $\hat{V}_N$, deduced from experiment or constructed from *ab initio* computation of electronic energy as a function of nuclear coordinates. For $\hat{T}_{VR}$, it is quite possible to use an exact form, derived from the simple Laplacian form in space-fixed cartesians, and this is the approach we wish to adopt in the variational calculation of rovibrational states of three- and four-atomic molecules. In section 2 we discuss how the desire for an $\hat{H}_{VR}$ that is separable, factorizable and amenable to a simple symmetry analysis affects the choice of a nuclear coordinate system (i.e. that subset of the molecular coordinates describing nuclear motion), starting with the separation of translation and then considering the choice of rovibrational coordinates. This choice determines crucially the detailed form of $\hat{T}_{VR}$, which in turn affects intimately the choice of basis set. We pay special attention to the choice of rotational coordinates via the embedding of a nuclear-fixed axis system.

Recently it was shown how to derive $\hat{T}_{VR}$ straightforwardly in any rovibrational coordinate system using a computer algebra program [1]. In section 3 we briefly review this method, with a clarification of the position of the nuclear-fixed origin. We have derived a new vibration–rotation kinetic energy operator which we propose to use in calculations on *sequentially bonded* four-atomic molecules such as C_2H_2 and

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H_2O_2; this is given in the Appendix. We have found that molecular symmetry (MS) groups [2–4] provide a rigorous and simple description of the symmetry of $\hat{T}_{VR}$, and using our new operator as an example, show how to use the MS group in the symmetry classification of coordinates and basis functions.

In section 4, we consider in detail how the coordinate system and the form of $\hat{T}_{VR}$ affect the choice of basis set, again using our new kinetic energy operator as an example. In choosing a basis set there are several criteria that may be of significance, including completeness of the basis, continuity of the basis functions, and the need for basis functions having finite local energy at, and finite integrals across, singular points. The simplicity with which the essential criteria can be satisfied determines how easily an efficient computer program for diagonalising $\hat{H}_{VR}$ can be designed.

2. Choosing a nuclear coordinate system

The $3N$ nuclear coordinates $\{r_i\}$ are best chosen to achieve as nearly as possible the two properties of *separability* and *factorizability* in the nuclear motion Hamiltonian $\hat{H}_N$. Separability means that the eigenfunctions of $\hat{H}_N$ are products of functions of non-intersecting subsets of $\{r_i\}$; this is most obviously the case when $\hat{H}_N$ can be written as a finite sum of independent operators

$$\begin{aligned}\hat{H}_N &= \hat{H}_{0,i}(r_1, \ldots, r_i) + \hat{H}_{i,j}(r_{i+1}, \ldots, r_j) \\ &\quad + \cdots + \hat{H}_{k,3N}(r_{k+1}, \ldots, r_{3N}) + \hat{H}'(r_1, \ldots, r_{3N}),\end{aligned} \tag{1}$$

where $\hat{H}'$ is small. Products of the eigenfunctions of the lower-dimensionality Hamiltonians $\hat{H}_{0,i}$, $\hat{H}_{i,j}$ etc. will be close to the true eigenfunctions of $\hat{H}_N$, and provide an excellent contracted basis for them. Exact separations with $\hat{H}' = 0$ are rare, although the motion of the nuclear centre of mass does separate exctly from the other nuclear motions. This may be proved in the following way. In cartesian coordinates referenced to a space-fixed (or laboratory-fixed) frame, the nuclear kinetic energy operator $\hat{T}_N$ has the simple form (using atomic units)

$$\begin{aligned}\hat{T}_N &= -\tfrac{1}{2}(\hat{T}_X + \hat{T}_Y + \hat{T}_Z) \\ &= -\frac{1}{2}\sum_{i=1}^{N}\frac{1}{M_i}\left(\frac{\partial^2}{\partial X_i^2} + \frac{\partial^2}{\partial Y_i^2} + \frac{\partial^2}{\partial Z_i^2}\right),\end{aligned} \tag{2}$$

where i is a nuclear label, and M_i a nuclear mass. Using the chain rule, $\hat{T}_X$ can be rewritten [5]

$$\hat{T}_X = \sum_{i=2}^{N}\frac{1}{M_i}\frac{\partial^2}{\partial(X_i - X_1)^2} + \frac{1}{M_1}\sum_{i,j=2}^{N}\frac{\partial^2}{\partial(X_i - X_1)\partial(X_j - X_1)} + \frac{1}{M_N}\frac{\partial^2}{\partial X_0^2}, \tag{3}$$

where X_0 is the X component of the position of the nuclear centre of mass and M_N the total nuclear mass

$$\begin{aligned}X_0 &= \frac{1}{M_N}\sum_{i=1}^{N} M_i X_i, \\ M_N &= \sum_{i=1}^{N} M_i.\end{aligned} \tag{4}$$

Obviously the same separation occurs in $\hat{T}_Y$ and $\hat{T}_Z$. The separation of (3) shows, applying the chain rule twice, that (2) can be rewritten as

$$\hat{T}_N = \hat{T}_{VR} - \frac{1}{2M_N}\left(\frac{\partial^2}{\partial X_0^2} + \frac{\partial^2}{\partial Y_0^2} + \frac{\partial^2}{\partial Z_0^2}\right), \tag{5}$$

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with [1]

$$\begin{aligned}\hat{T}_{\mathrm{VR}} &= -\frac{1}{2}\sum_{j,k=1}^{3N-3}\sum_{R\in\{X,Y,Z\}}\sum_{i=1}^{N}\frac{1}{M_i}\frac{\partial t_j}{\partial R_i}\frac{\partial t_k}{\partial R_i}\frac{\partial^2}{\partial t_j \partial t_k} \\ &\quad -\frac{1}{2}\sum_{j=1}^{3N-3}\sum_{R\in\{X,Y,Z\}}\sum_{i=1}^{N}\frac{1}{M_i}\frac{\partial^2 t_j}{\partial R_i^2}\frac{\partial}{\partial t_j} \\ &\equiv -\frac{1}{2}\left(\sum_{j,k} g^{jk}\frac{\partial^2}{\partial t_j \partial t_k}+\sum_j h^j\frac{\partial}{\partial t_j}\right), \end{aligned} \tag{6}$$

where $\{t_i\}$ is a set of $3N-3$ independent functions of projections of internuclear distances onto space-fixed axes; the separation of (3) (and (5)) makes it clear that this is the correct definition of a valid *rovibrational* coordinate system. The choice of the $\{t_i\}$ will determine the extent to which the rovibrational Hamiltonian $\hat{H}_{\mathrm{VR}} = \hat{T}_{\mathrm{VR}} + \hat{V}_{\mathrm{N}}$ can be made approximately separable. We should note, though, that the interesting resonances in molecular spectra generally involve coupling of two or more motions that are inseparable in any coordinate system.

$\hat{H}_{\mathrm{VR}}$ is factorizable if it can be written as a sum of products of one-dimensional operators

$$\hat{H}_{\mathrm{VR}} = \sum_i \prod_{j=1}^{3N-3}\hat{O}_{ij}(t_j). \tag{7}$$

Such a form is computationally desirable because it means that the matrix elements of $\hat{H}_{\mathrm{VR}}$ can be constructed purely from one-dimensional integrals, which are cheap to calculate: there will usually be a Gauss quadrature rule to evaluate them exactly [6], with no need for complicated analytic formulae (and possible attendant problems of numerical instability). There are methods that diagonalize $\hat{H}_{\mathrm{VR}}$ approximately, such as the discrete variable representation (DVR) developed by Light and coworkers [7–9], and the collocation method of Yang and Peet [10], which respectively partially or completely avoid the evaluation of multidimensional integrals; but in strictly variational methods, the explicit evaluation of matrix elements of the Hamiltonian cannot be avoided, and factorizability remains a desirable property of $\hat{H}_{\mathrm{VR}}$.

2.1. *Vibrational coordinates*

The vibrational coordinates $\{q_i\}$ may be defined to be $3N-6$ independent rotationally invariant functions (i.e. invariant under rotations of the N nuclei about axes fixed in space) of any valid set of $3N-3$ rovibrational coordinates $\{t_i\}$ (see above). It is usual to consider the $\{q_i\}$ in relation to a nuclear-fixed frame (otherwise known as body-fixed or molecule-fixed), a set of cartesian axes that rotate with the molecule in the sense that the operation of rotating all the N nuclei through a particular angle about a particular axis has an identical effect on the frame [11]. (The impossibility of defining a nuclear-fixed frame when all the nuclei are collinear will not be considered until sections 3.2 and 4.) It follows that the $\{q_i\}$ can be constructed as functions of the nuclear-fixed cartesian coordinates. Furthermore, since they must also satisfy the definition of valid rovibrational coordinates, i.e. be functions of projections of internuclear distances onto space-fixed axes, the $\{q_i\}$ can always be written as functions (linear functions in the case of normal coordinates) of projections of internuclear distances onto nuclear-fixed axes. The fact that $\hat{V}_{\mathrm{N}}$ is invariant to rotations and is not a differential operator means that it can be expressed as a function purely of the $\{q_i\}$. In general, however, for a particular nuclear configuration the values of the $\{q_i\}$ depend on the way the nuclear-fixed frame has been attached to the nuclei; and so it is more natural to express $\hat{V}_{\mathrm{N}}$ in terms of *internal* coordinates,

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meaning a set $\{q_i\}$ having the special property of being functions purely of the actual internuclear distances, and therefore independent of the concept of the nuclear-fixed frame.

Normal coordinates are an obvious starting point in seeking separability in the vibrational problem since they allow both the harmonic approximation of $\hat{V}_N$ and the rigid-rotor approximation of $\hat{T}_{VR}$ to separate exactly into sums of one-dimensional operators. Watson [12, 13] has obtained a rather simple but *exact* form for $\hat{T}_{VR}$ in terms of vibrational normal coordinates $\{Q_i\}$ (plus the three components of rovibrational angular momentum $\hat{J}$ about nuclear-fixed axes $\{J_x, J_y, J_z\}$); its dominant part is a sum of one-dimensional operators. It is, rather, the (often large) anharmonic terms in $\hat{V}_N$, strongly coupling the $\{Q_i\}$ at non-infinitesimal vibrational amplitudes, that drastically reduce the separability of $\hat{H}_{VR}$ in normal coordinates. Moreover, Watson's $\hat{T}_{VR}$ has the disadvantage that the inverse inertia matrix elements that occur in it, coupling together all the rovibrational motions, are not factorizable except in an infinite expansion [12]. There are calculations of rovibrational energies of polyatomic molecules using the exact Watson Hamiltonian (see, for example, [14–16]), but they require the costly direct evaluation of multidimensional integrals.

$\hat{V}_N$ is commonly presented in a factorizable form: for example, as an approximate analytic fit to a set of *ab initio* electronic energy points, or as an expansion in a truncated Taylor series about the equilibrium geometry. The availability of analytic *ab initio* derivatives of $\hat{V}_N$ up to fourth order in nuclear-fixed cartesian displacement coordinates [17–20] makes the latter option rather inexpensive and provides at least a reasonable starting point for the further refinement of $\hat{V}_N$. A Taylor expansion in cartesian displacement coordinates is easily transformed to normal coordinates or internal coordinates. *Valence* internal coordinates, i.e. a well chosen set of bond lengths, bending and torsion angles, are particularly suitable for the expansion of $\hat{V}_N$ in the vicinity of the equilibrium geometry because they reflect the equilibrium bonding pattern. The principal advantage of valence coordinates over normal coordinates is that the Taylor expansion of $\hat{V}_N$ should have a much greater radius of convergence in well chosen valence coordinates: they are curvilinear and so describe the curved nuclear trajectories of large amplitude bending and torsional vibration much more closely. In the case of stretches, the error arising from truncation of the Taylor expansion can often be reduced by a further transformation to Morse or SPF [21] functions of the valence coordinates. These transformations can equally well be regarded as improvements to the separability and compact factorizability of $\hat{V}_N$.

Following work of Sutcliffe and others (see the reviews [22–24]), Handy has shown [1], on the basis of equation (6), how to derive an *exact* $\hat{T}_{VR}$ straightforwardly for, in principle, any number of nuclei and any choice of vibrational coordinates $\{q_i\}$. He has given a number of examples, using internal coordinates, for molecules with three or four nuclei; all the forms of $\hat{T}_{VR}$ given in [1], some of which had been derived previously by other methods, are factorizable. The method of [1] is discussed in section 3.1.

Our experience is that the $\hat{T}_{VR}$ in valence coordinates are factorizable so long as simple lengths and angles are used; linear combinations of lengths and angles, however, lead to unfactorizable terms. This is a reason why, for example, it is difficult to design a good set of vibrational coordinates for ammonia (NH_3). To begin with, the simple set of three bond lengths and three bond angles does not span the full vibrational space since it cannot describe the umbrella motion. A more sophisticated attempt to exploit the symmetry by using symmetry combinations of lengths and angles leads to an unfactorizable $\hat{T}_{VR}$. It may be possible in the case of ammonia to

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use coordinates designed for formaldehyde (H_2CO) (see [1]), but they do not treat the three hydrogen nuclei in a symmetric manner. It is to be expected that only coordinates adapted as closely as possible to the symmetry of the true molecular Hamiltonian will facilitate efficient calculation (see sections 2.2 and 3.2).

For the sake of completeness, we note that many other vibrational coordinate systems appear in the literature, perhaps most notably the Jacobi [25, 26], Radau [25], and hyperspherical [27–29] internal coordinates for three-atomic molecules. These three examples are all orthogonal coordinate systems, and as a result process the property that $\hat{T}_{VR}$ contains no mixed second derivatives, which is a form of separability. Jacobi coordinates have been shown to be particularly suitable for the isomerization process A–B–C ↔ A–C–B [30]; the symmetry properties of hyperspherical coordinates make them attractive for homonuclear three-atomic molecules [31–33]. Although $\hat{T}_{VR}$ is more separable (and has much simpler forms) in these coordinate systems than in valence coordinates, it is possible to argue that $\hat{V}_N$ will be more separable in valence coordinates, at least in regions of the surface where bond-breaking processes are not occurring.

2.2. *Rotational coordinates*

When using internal coordinates for vibration, it is possible (see section 3.1), in the method of [1], to derive $\hat{T}_V$, that part of $\hat{T}_{VR}$ containing only vibrational coordinates, without any consideration of rotational coordinates, and the resulting operator may be used to calculate the $J = 0$ rovibrational states. Otherwise, one requires the remainder of $\hat{T}_{VR}$, $\hat{T}_R$, and so it is necessary to make an explicit choice of rotational coordinates, of which there are three when they are chosen to be independent. The rotational coordinates may be defined to be a parameterization of the operator that rotates the nuclei from a position where the nuclear-fixed and space-fixed frames are coincident, to their actual position. Although there exist other parameterizations of rotation [34], the usual choice is to use three Euler angles. Superficially then, different methods use the same rotational coordinates, the Euler angles, but in fact their meaning depends drastically on how the directions of the rotating axes are defined with respect to the nuclei.

Before discussing the orientation of the nuclear-fixed frame, we must consider its origin. The position of the origin is important because it determines the meaning of the rovibrational angular momentum $\hat{\boldsymbol{J}}$. J is a good quantum number when referred to *any* origin, because all rotations about space-fixed axes through all origins are true symmetry operations of the molecular Hamiltonian. In spectroscopy, however, J is used to label eigenfunctions of angular momentum about the *molecular* centre of mass (i.e. including the electrons); so if we wish to compare directly calculations with experiment, the origin *must* be placed here. The exact separation of translation of the *nuclear* centre of mass in $\hat{T}_N$ ((3) and (5)), however, means that when $\hat{H}_{VR}$ is written down in any valid set (as defined above) of $3N - 3$ rovibrational coordinates $\{t_i\}$, the origin of the nuclear-fixed frame has implicitly been placed there. This is because the $\partial/\partial t_i$ in the $\hat{T}_{VR}$ of (6) operate implicitly at *fixed* $\boldsymbol{X}_0$, the position vector of the nuclear centre of mass in the space-fixed frame, and so rotation is about this point. The fact that the nuclear and molecular centres of mass are not coincident leads to tiny errors of a similar magnitude to those arising from the Born–Oppenheimer approximation; although these can be partially corrected by using atomic instead of nuclear masses in calculations (see, for example, [35]). The position of the nuclear-fixed origin was dealt with misleadingly in [1]; in section 3.1 we clarify the role of the origin in the derivation of $\hat{T}_{VR}$.

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If normal coordinates are used for vibration, it is natural to orient the nuclear-fixed frame according to the rotational Eckart conditions [36, 4]. The reason for this is as follows. When normal coordinates are determined by diagonalizing the matrix of second derivatives of $\hat{V}_N$ with respect to the $3N$ mass-weighted nuclear-fixed cartesian displacement coordinates, there are (for a non-linear equilibrium geometry) six zero-frequency eigenvectors in addition to the $3N - 6$ vibrational normal coordinates $\{Q_i\}$. It is possible to show [37], using the invariance of $\hat{V}_N$ to translations and rotations, that three of the zero-frequency eigenvectors are translational coordinates $\{T_x, T_y, T_z\}$ which describe the position of the nuclear centre of mass in the nuclear-fixed frame, and that three are the infinitesimal rotation (about nuclear-fixed axes) coordinates $\{R_x, R_y, R_z\}$. Therefore pure motion along a normal coordinate requires, if it is a vibrational coordinate, that the frame should be embedded so that the $\{T_x, T_y, T_z\}$ and $\{R_x, R_y, R_z\}$ are all zero. The conditions for this are precisely the siting of the origin at the nuclear centre of mass, as established above; and the orientation of the axes, with respect to the nuclei, according to the rotational Eckart conditions.

Nuclear-fixed frames that differ in the way that the axes are defined with respect to the nuclei differ consequently in the amount of vibrational angular momentum that can occur, or, equivalently, how well vibration and rotation are separated. In the Eckart frame, this vibrational angular momentum is kept to a minimum, but not removed altogether [4]. It is possible though to define frames (for example the principal axis frame [4]) in which ordinary small-amplitude vibration corresponds to large swings of the axis system, and therefore large angular momentum. There must be some residual vibrational angular momentum in *any* frame because vibration and rotation in a non-rigid system are intrinsically coupled. An example of this is provided by symmetric tops, in which there is always some *l*-type doubling (a form of vibration–rotation coupling) to prevent K from being a perfect quantum number. K is not a good quantum number because it labels irreducible representations of $\mathbf{D}_\infty$, a subgroup of the **K**(mol) group [4] (all rotations of the molecule about all nuclear-fixed axes through its centre of mass), and these groups are near but not true symmetry groups of the molecular Hamiltonian.

In the simplest application of the method of [1] for deriving $\hat{T}_{VR}$ (see section 3.1), the Eckart frame is not used. Of course $\hat{H}_{VR}$ has the same eigenvalues whatever nuclear-fixed frame is used to derive $\hat{T}_{VR}$, because any derivation starts out from the undeniably correct form that $\hat{T}_{VR}$ has in space-fixed cartesians, and no approximations are made in the successive coordinate transformations. Moreover, there is a simple relationship, for a given eigenfunction of $\hat{H}_{VR}$, between the two functional forms taken by its vibrational parts with respect to two different definitions of the nuclear-fixed axis directions. This may be shown as follows.

Any eigenfunction of $\hat{H}_{VR}$ has the general form [11]

$$\Psi^J_M = \sum_K \bar{D}^J_{MK}(\Omega_1)^* \psi^{(1)}_K(\boldsymbol{q}), \tag{8}$$

where (Ω_1) represents the three Euler angles that specify the orientation of nuclear-fixed frame 1 with respect to the space-fixed frame, $\boldsymbol{q}$ are the $3N - 6$ vibrational coordinates, and $\boldsymbol{D}^J$ is a Wigner rotation matrix [38], with

$$\bar{\boldsymbol{D}}^J = \frac{1}{2\pi}\left(\frac{2J+1}{2}\right)^{1/2} \boldsymbol{D}^J. \tag{9}$$

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Equivalently, replacing frame 1 by frame 2, we can write

$$\Psi_M^J = \sum_K \bar{D}_{MK}^J(\Omega_2)^* \psi_K^{(2)}(\boldsymbol{q}). \tag{10}$$

We seek the relationship between the $\{\psi_K^{(1)}\}$ and the $\{\psi_K^{(2)}\}$. Consider first the rotation operator identify

$$\hat{\Omega}_2 = \hat{\Omega}_1(\hat{\Omega}_1^{-1}\hat{\Omega}_2). \tag{11}$$

Let $\hat{\Omega}_i$ (where $i = 1$ or 2) be the operator that rotates the N nuclei from a position where frame i is coincident with the space-fixed frame to a position where the orientation of frame i with respect to the space-fixed frame is specified by Euler angles (Ω_i); therefore the rotations $\hat{\Omega}_1$ and $\hat{\Omega}_2$ are parameterized by Euler angles (Ω_1) and (Ω_2) respectively, with respect to the space-fixed frame. Now $\hat{\Omega}_2\hat{\Omega}_1^{-1}$ is an operator whose effect is to rotate frame 1 onto the initial location of frame 2 through an angle δ about an axis $\hat{\boldsymbol{n}}_1$ *fixed in frame 1*, where δ is invariant under a change in the values of (Ω_1) and (Ω_2) at fixed $\boldsymbol{q}$ and is therefore a function purely of $\boldsymbol{q}$. The operator $\hat{\Omega}_1^{-1}\hat{\Omega}_2$, on the other hand, is a rotation of the nuclei through the same angle δ, but about $\hat{\boldsymbol{n}}$, the corresponding axis *fixed in the space-fixed frame*; $\hat{\Omega}_1^{-1}\hat{\Omega}_2$ is, therefore, parameterized by Euler angles $(\Omega_{12}(\boldsymbol{q}))$, with respect to the space-fixed frame, that are functions purely of vibrational coordinates. The identity (11) thus has the matrix representation

$$D_{MK}^J(\Omega_2) = \sum_{K'} D_{MK'}^J(\Omega_1) D_{K'K}^J(\Omega_{12}(\boldsymbol{q})) \tag{12}$$

or

$$\bar{D}_{MK}^J(\Omega_2)^* = \sum_{K'} \bar{D}_{MK'}^J(\Omega_1)^* D_{K'K}^J(\Omega_{12}(\boldsymbol{q}))^*. \tag{13}$$

Substituting (13) into (10), we obtain

$$\Psi_M^J = \sum_{K'} \bar{D}_{MK'}^J(\Omega_1)^* \left(\sum_K D_{K'K}^J(\Omega_{12}(\boldsymbol{q}))^* \psi_K^{(2)}(\boldsymbol{q}) \right) \tag{14}$$

and, by comparison with (8), the identification

$$\psi_K^{(1)}(\boldsymbol{q}) = \sum_{K'} D_{KK'}^J(\Omega_{12}(\boldsymbol{q}))^* \psi_{K'}^{(2)}(\boldsymbol{q}). \tag{15}$$

Although simple, (15) is a potentially powerful relation: having calculated representations of the $\{\Psi_M^J\}$ using one particular definition of the nuclear-fixed frame, one could use (15), at limited extra computational cost, to obtain the corresponding representations for any other definition of the frame.

The relatively poorer vibration–rotation separation in a non-Eckart frame means that when using the symmetric top functions $\bar{D}_{MK}^{J*} \equiv |J, K, M\rangle$ as the rotational basis in calculations on linear molecules and symmetric tops, K is a poorer quantum number than usual, and it is important to allow full mixing (as in (8) and (10)) between the different K, and probably also to obtain a different optimized vibrational basis for each $|K|$ (see section 4.3). For asymmetric tops, this would have to be done however the nuclear-fixed frame were defined. There remains also the following conundrum. Components of $\hat{\boldsymbol{J}}$ are introduced via formulae [38] relating them to Euler angle derivatives; how then can J have the same meaning (as it must) for two different embeddings of the nuclear-fixed frame, where the Euler angles have a different meaning in each case? In particular, consider a molecule with N nuclei, the directions of the nuclear-fixed axes being defined by only three of the nuclei. How can the

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angular momentum of the other $N - 3$ nuclei enter into J? One explanation lies in noting that a derivative with respect to an Euler angle is a partial derivative for which the values of all of the other $3N - 4$ rovibrational coordinates are implicitly fixed. Suppose, for simplicitly, that vibration is described by internal coordinates, dependent only on internuclear distances, and that they are all fixed while an Euler angle is varied. Even though $N - 3$ of the nuclei play no part in the definition of the Euler angles, the variation of one of these causes *all* the nuclei to rotate around their centre of mass: $\hat{\boldsymbol{J}}$ measures the angular momentum of the whole, frozen, nuclear framework. This result is proved mathematically in [11].

In the examples of $\hat{T}_{\mathrm{VR}}$ in [1], nuclear-fixed axis directions are related in simple ways to the internuclear vectors, but never in a way adapted to the underlying symmetry of the molecular Hamiltonian. This is not a practical approach, because it prevents the use of symmetry when considering rotational coordinates separately from vibrational coordinates. If axis directions are chosen more carefully, so that the equivalent rotations [4] of the elements of the MS group do not depend on any vibrational coordinates, then rotational coordinates have well defined symmetry separate from vibrational coordinates. This is discussed further, with examples, in section 3.2.

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3. The kinetic energy operator $\hat{T}_{\mathrm{VR}}$

3.1. *Derivation of* $\hat{T}_{\mathrm{VR}}$

The derivation of $\hat{T}_{\mathrm{VR}}$ in terms of vibrational and rotational coordinates, as defined in section 2, is a subject that has received much attention; see, for example, [1, 12, 13, 22–24, 39, 40] and references therein. The importance of the problem has recently been reemphasized by Chapuisat *et al.* [40]; in a series of papers they have systematically studied several alternative approaches. They observe [40] that 'Handy's method [1] is clearly efficient for numerical calculations'. In this section, we shall briefly review and clarify this method.

In [1], the derivation of $\hat{T}_{\mathrm{VR}}$ begins naturally from the simple form (2) in space-fixed cartesians. This expression is transformed, using a computer algebra program, into the final coordinates $\{t_i\}$, which are $3N - 6$ vibrational coordinates and three Euler angles, according to (6). It is necessary to split the transformation into simple stages, in each of which the chain rule is used to transform the differential operators, in order to limit the complexity of the intermediate algebra. The first stage, for example, is the transformation from the $3N$ space-fixed Cartesians to $3N - 6$ elementary vibrational coordinates (e.g., all the internuclear distances) plus three elementary rotational coordinates (suitable for later transformation into the Euler angles). Since these $3N - 3$ elementary rovibrational coordinates from a valid set, in the sense of depending only on internuclear distances and their projections onto space-fixed axes (see section 2), then, in this first stage, the translation of the nuclear centre of mass has been implicitly separated. Subsequent stages of the derivation form a series of simple coordinate transformations designed to arrive eventually at the final coordinates. As a final step, derivatives with respect to Euler angles are replaced by angular momentum operators (see below).

In [1], it was suggested that the origin of the nuclear-fixed frame could be placed at one of the nuclei, whereas, as argued in section 2.2, when using a valid set of rovibrational coordinates, the origin is at the nuclear centre of mass. We will show that the equations by which the origin is introduced in [1] are in fact independent of

the origin, so that there is never any need to explicitly consider the position of the origin at all. These equations are

$$\boldsymbol{X}_i = \boldsymbol{S}(\alpha, \beta, \gamma)\boldsymbol{x}_i(\boldsymbol{q}) + \boldsymbol{X}_0, \qquad i = 1 \ldots N, \tag{16}$$

where $\boldsymbol{X}_i$ is the position vector of nucleus i and $\boldsymbol{X}_0$ the position vector of the nuclear-fixed origin, both in space-fixed cartesians; $\boldsymbol{S}$ is a direction cosine matrix whose elements are trigonometric functions of the Euler angles α, β, γ (called correspondingly ϕ, θ, χ in [38]); and $\boldsymbol{x}_i$ is the position vector of nucleus i in nuclear-fixed cartesians, expressed in terms of the vibrational coordinates $\boldsymbol{q}$.

Consider the four-nuclear example. Let $\boldsymbol{X}_0$ by an origin fixed with respect to the nuclei in the sense that the operation of translating all the nuclei a particular distance in a particular direction has an identical effect on the origin; it can then be written as a weighting

$$\boldsymbol{X}_0 = (c_1\ c_2\ c_3\ c_4)\begin{pmatrix}\boldsymbol{X}_1\\ \boldsymbol{X}_2\\ \boldsymbol{X}_3\\ \boldsymbol{X}_4\end{pmatrix}, \tag{17}$$

with

$$c_1 + c_2 + c_3 + c_4 = 1. \tag{18}$$

Then (16) becomes

$$-\boldsymbol{S}\begin{pmatrix}\boldsymbol{x}_1\\ \boldsymbol{x}_2\\ \boldsymbol{x}_3\\ \boldsymbol{x}_4\end{pmatrix} = \begin{pmatrix}c_1 - 1 & c_2 & c_3 & c_4\\ c_1 & c_2 - 1 & c_3 & c_4\\ c_1 & c_2 & c_3 - 1 & c_4\\ c_1 & c_2 & c_3 & c_4 - 1\end{pmatrix}\begin{pmatrix}\boldsymbol{X}_1\\ \boldsymbol{X}_2\\ \boldsymbol{X}_3\\ \boldsymbol{X}_4\end{pmatrix}. \tag{19}$$

Now define the non-singular matrices $\boldsymbol{A}$ and $\boldsymbol{B}$

$$\boldsymbol{A} = \begin{pmatrix}1 & -1 & 0 & 0\\ 1 & 0 & -1 & 0\\ 1 & 0 & 0 & -1\\ c_1 & c_2 & c_3 & c_4\end{pmatrix}, \qquad \boldsymbol{B} = \begin{pmatrix}1 & -1 & 0 & 0\\ 1 & 0 & -1 & 0\\ 1 & 0 & 0 & -1\\ 1 & 1 & 1 & 1\end{pmatrix}, \tag{20}$$

and rewrite (19) as

$$-\boldsymbol{SA}\begin{pmatrix}\boldsymbol{x}_1\\ \boldsymbol{x}_2\\ \boldsymbol{x}_3\\ \boldsymbol{x}_4\end{pmatrix} = \boldsymbol{A}\begin{pmatrix}c_1 - 1 & c_2 & c_3 & c_4\\ c_1 & c_2 - 1 & c_3 & c_4\\ c_1 & c_2 & c_3 - 1 & c_4\\ c_1 & c_2 & c_3 & c_4 - 1\end{pmatrix}\boldsymbol{B}^{-1}\boldsymbol{B}\begin{pmatrix}\boldsymbol{X}_1\\ \boldsymbol{X}_2\\ \boldsymbol{X}_3\\ \boldsymbol{X}_4\end{pmatrix}, \tag{21}$$

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which, using (18), is the same as

$$-\boldsymbol{S}\begin{pmatrix} \boldsymbol{x}_1 - \boldsymbol{x}_2 \\ \boldsymbol{x}_1 - \boldsymbol{x}_3 \\ \boldsymbol{x}_1 - \boldsymbol{x}_4 \\ c_1\boldsymbol{x}_1 + c_2\boldsymbol{x}_2 + c_3\boldsymbol{x}_3 + c_4\boldsymbol{x}_4 \end{pmatrix} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \times \begin{pmatrix} \boldsymbol{X}_1 - \boldsymbol{X}_2 \\ \boldsymbol{X}_1 - \boldsymbol{X}_3 \\ \boldsymbol{X}_1 - \boldsymbol{X}_4 \\ \boldsymbol{X}_1 + \boldsymbol{X}_2 + \boldsymbol{X}_3 + \boldsymbol{X}_4 \end{pmatrix} \tag{22}$$

or

$$\boldsymbol{S}(\alpha, \beta, \gamma)\begin{pmatrix} \boldsymbol{x}_1 - \boldsymbol{x}_2 \\ \boldsymbol{x}_1 - \boldsymbol{x}_3 \\ \boldsymbol{x}_1 - \boldsymbol{x}_4 \end{pmatrix}(\boldsymbol{q}) = \begin{pmatrix} \boldsymbol{X}_1 - \boldsymbol{X}_2 \\ \boldsymbol{X}_1 - \boldsymbol{X}_3 \\ \boldsymbol{X}_1 - \boldsymbol{X}_4 \end{pmatrix}; \tag{23}$$

$c_1\boldsymbol{x}_1 + c_2\boldsymbol{x}_2 + c_3\boldsymbol{x}_3 + c_4\boldsymbol{x}_4 = 0$ as it is the origin. Compared with (16), we see that having removed three redundancies, (23) are purely in terms of the projections of interparticle distances onto nuclear- and space-fixed axes, independent of the nuclear-fixed origin, which, as stated above, need not be considered in the derivation. The generalization of (23) to N nuclei is obvious.

Equations (23) can be rearranged to obtain expressions for introducing the Euler angles in terms of the $\{\boldsymbol{X}_1 - \boldsymbol{X}_i\}$ and $\{q_i\}$; they are also used to eliminate the $\{\boldsymbol{X}_i\}$ from $\hat{T}_{\mathrm{VR}}$. *They contain all the information about how the nuclear-fixed frame is embedded*: for different frames, the dependence of the $\{\boldsymbol{x}_1 - \boldsymbol{x}_i\}$ on the $\{q_i\}$ is different. As long as the frame definition is fairly simple, it is easy to write down expressions for the $\{\boldsymbol{x}_1 - \boldsymbol{x}_i\}$ in terms of the $\{q_i\}$ by inspection of a diagram or model.

Having arrived at a form for $\hat{T}_{\mathrm{VR}}$ in terms of $3N - 6$ vibrational coordinates, three Euler angles, and derivatives with respect to them, the derivatives with respect to Euler angles are replaced, using standard relations [38], by the $\{\hat{J}_x, \hat{J}_y, \hat{J}_z\}$ operators (rovibrational angular momentum about nuclear-fixed axes through the nuclear centre of mass (see section 2.2), obeying anomalous commutation relations [38]). If one is interested only in states with $J = 0$, then all the terms in $\hat{T}_{\mathrm{VR}}$ containing $\{\hat{J}_x, \hat{J}_y, \hat{J}_z\}$ may be set to zero. The resulting $\hat{T}_{\mathrm{V}}$ operator can be obtained more simply, in the case where the $\{q_i\}$ are internal coordinates, as follows. If, in the first stage of the derivation, the transformation is made from $3N$ space-fixed cartesians to just $3N - 6$ $\{q_i\}$, then derivatives with respect to Euler angles have been omitted from $\hat{T}_{\mathrm{VR}}$, which is the same as implicitly setting all the $\{\hat{J}_x, \hat{J}_y, \hat{J}_z\}$ to zero. This derivation of $\hat{T}_{\mathrm{V}}$ in internal coordinates makes no use of equations (23) (or (16)), and so $\hat{T}_{\mathrm{V}}$ is independent of the definition of the nuclear-fixed frame.

We should note that throughout the present work, $\hat{\boldsymbol{J}}$ represents the rovibrational angular momentum. In molecules where there is electronic angular momentum, either orbital ($\hat{\boldsymbol{L}}$) or spin ($\hat{\boldsymbol{S}}$) or both, $\hat{\boldsymbol{J}}$ properly refers to the total rovibronic angular momentum, and so the $\hat{\boldsymbol{J}}$ that appears in our expressions should be replaced by $\hat{\boldsymbol{J}} - \hat{\boldsymbol{L}} - \hat{\boldsymbol{S}}$.

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Matrix elements of $\hat{T}_{VR}$ (using the form (6)) are evaluated as

$$\langle\Phi|\hat{T}_{VR}|\Phi'\rangle = \int dt_1 \ldots dt_{3N-3}\, s\Phi^*\hat{T}_{VR}\Phi', \tag{24}$$

where the weight function s is the Jacobian for the transformation from $\{\boldsymbol{X}_i, i = 1, \ldots, N\}$ to $\{t_i, i = 1, \ldots, 3N-3; \boldsymbol{X}_0\}$. In principle it would be straightforward to determine s algebraically using (23), as the $(3N-3)$-square determinant

$$\begin{aligned} s &= \left|\frac{\partial(\boldsymbol{X}_i - \boldsymbol{X}_1)}{\partial t_j}\right| \\ &\equiv \left(|g^{jk}| \prod_{i=1}^{N} M_i^3\right)^{-1/2} \end{aligned} \tag{25}$$

where g^{jk} is defined in (6). In fact it is easier simply to identify s as a weight function that makes $\hat{T}_{VR}$ Hermitian [24]; for instance it must contain the factor $\sin\beta$ so that $\hat{T}_{VR}$ is Hermitian when integrated over the Euler angles. It is common practise to absorb s (or part of it) into $\hat{T}_{VR}$ and basis functions Φ so that the latter are transformed as [1, 37]

$$\begin{aligned} \overline{\hat{T}_{VR}} &= s^{1/2}\hat{T}_{VR}s^{-1/2}, \\ \bar{\Phi} &= s^{1/2}\Phi; \end{aligned} \tag{26}$$

matrix elements are evaluated as

$$\langle\Phi|\hat{T}_{VR}|\Phi'\rangle = \int dt_1 \ldots dt_{3N-3}\, \bar{\Phi}^*\, \overline{\hat{T}_{VR}}\, \overline{\Phi'}, \tag{27}$$

and, writing $a = s^{-1/2}$, form (6) becomes

$$\begin{aligned} \overline{\hat{T}_{VR}} = -\frac{1}{2}\Bigg(&\sum_{j,k} g^{jk}\frac{\partial^2}{\partial t_j \partial t_k} + \sum_j h^j \frac{\partial}{\partial t_j} \\ &+ \frac{2}{a}\sum_{j,k} g^{jk}\frac{\partial a}{\partial t_j}\frac{\partial}{\partial t_k} \\ &+ \frac{1}{a}\sum_{j,k} g^{jk}\frac{\partial^2 a}{\partial t_j \partial t_k} + \frac{1}{a}\sum_j h^j \frac{\partial a}{\partial t_j}\Bigg). \end{aligned} \tag{28}$$

(The transformation has no effect on $\hat{V}_N$.) The appearance of non-differential terms in a T_{VR} operator (equation (28), third row) shows that a transformation to alter the weight function for integration has been made. As has been previously noted [39, 40], this is the origin of the Watson U term [12].

We have used the method of [1] to derive a new $\hat{T}_{VR}$ (see Appendix) for four-atomic molecules in the *sequential bonding* coordinates r_1, r_2, r_3, θ_1, θ_2, ϕ, shown, together with their ranges, in figure 1; and with the nuclear-fixed frame (b) shown in figure 3. The purely vibrational parts $\hat{T}_V$ of the new operator are identical, having used the same vibrational coordinates, to those in the form presented in Appendix B of [1] (apart from a typographical error there in the $\partial^2/\partial\phi\,\partial\theta_1$ and $\partial^2/\partial\phi\,\partial\theta_2$ terms); the parts containing $\{\hat{J}_x, \hat{J}_y, \hat{J}_z\}$ differ because of the different frame definitions (see section 3.2), and it is these ($\hat{T}_R$) that are new. The Jacobian s for the transformation from $\{\boldsymbol{X}_i, i = 1, \ldots, N\}$ to sequential bonding coordinates and Euler angles (and $\boldsymbol{X}_0$) is $r_1^2 r_2^2 r_3^2 \sin\theta_1 \sin\theta_2 \sin\beta$. In the derivation of our new operator and that in Appendix B of [1], the $r_1^2 r_2^2 r_3^2$ factor was absorbed into $\hat{T}_{VR}$ and the stretching basis

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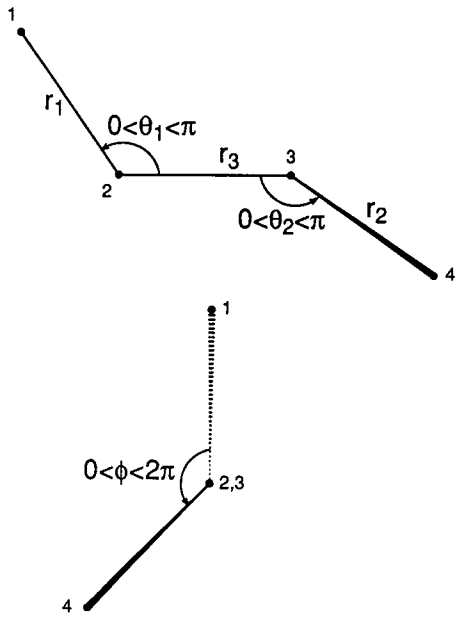


Figure 1. Sequential bonding coordinates and nuclear labels for a four-atomic molecule.

functions, as described above; the purpose of this is to ensure that harmonic oscillator and Morse functions are good stretching basis functions.

3.2. *The symmetry of* $\hat{T}_{VR}$

In practical calculations it is most efficient to use a basis set consisting of functions adapted to the symmetry of $\hat{H}_{VR}$, allowing several small diagonalizations instead of one large one. The symmetry classification must be carried out using a group of true symmetry operations of $\hat{H}_{VR}$, otherwise the blocking of the $\hat{H}_{VR}$ matrix will be only approximate. It is, therefore, an error to use the molecular point group or the molecular rotation group, which are only near symmetry groups. It is best to classify rovibrational basis functions using the molecular symmetry (MS) group [2–4], a true symmetry group of $\hat{H}_{VR}$ consisting of permutations of identical nuclei and their products with E^*, the inversion with respect to the space-fixed frame.

Watson's $\hat{T}_{VR}$ [12, 13] is derived (i) in normal coordinates, which depend on the detailed shape of $\hat{V}_n$ around the equilibrium point; and (ii) using the rotational Eckart conditions, which depend upon its geometry. The choice of all the rovibrational coordinates thus depends on $\hat{V}_N$, and since it is not possible, in the normal way, to embed a nuclear-fixed frame when all the nuclei are collinear, it is no surprise that a linear equilibrium geometry has to be treated as a special case [13]. If, however, one employs rovibrational coordinates *independent of* $\hat{V}_N$, then $\hat{T}_{VR}$ is independent of the molecular equilibrium geometry (suggesting immediately that the symmetries of $\hat{T}_{VR}$ do not belong to the point group, whose determination follows from the equilibrium geometry.) For example, the sequential bonding coordinates are suitable for any four-atomic molecule with the bonding pattern A–B–C–D (irrespective of whether or not the equilibrium geometry is linear; the possibility of collinearity of the nuclei does however give rise to singularities in $\hat{T}_{VR}$ (see section 4)). Consider H_2O_2 and C_2H_2: these two molecules have quite different molecular point groups, as appropriate to

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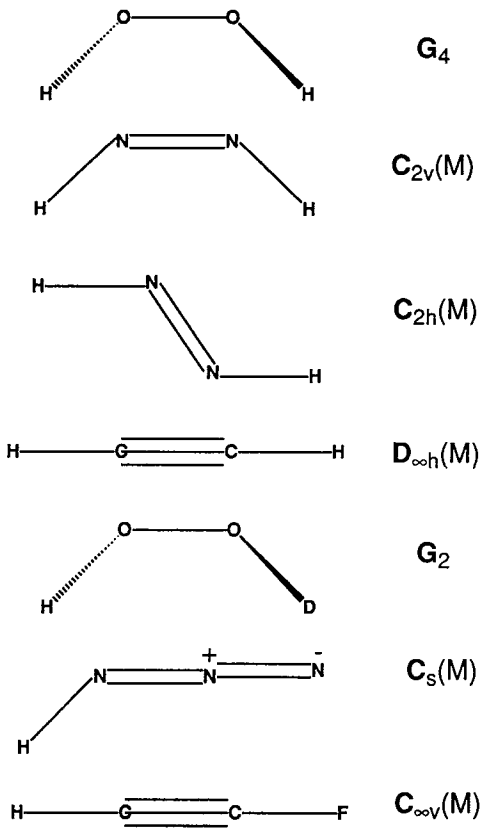


Figure 2. The nomenclature of MS groups for four-atomic molecules (including torsional tunnelling) according to equilibrium geometry.

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their different equilibrium geometries, but the same $\hat{T}_{VR}$ (assuming that nuclear-fixed axis directions are chosen in the same way). If the form of $\hat{V}_N$ is chosen carefully (in this example $\hat{V}_N$ must be expanded in $\cos\phi$ instead of $(\phi - \phi_e)$—also necessary for studying the torsional tunelling splitting in H_2O_2 [41]), then for each molecule the form used for $\hat{H}_{VR}$ has the same symmetries and the same MS group can be used; the symmetry analysis is identical in the two cases and the same computer program can be used for calculating the rovibrational states of the two molecules. There are in fact just two MS groups for sequentially bonded four-atomic molecules: one for A–B–B–A molecules and one for those of type A–B–B–B, A–B–B–C, A–B–C–A, A–B–C–C, or A–B–C–D. By convention, the names by which the groups and their irreducible representations are known vary according to the equilibrium geometry of the molecules [4]: H_2O_2 and C_2H_2 have the same MS group but it is called $\mathbf{G}_4$ or $\mathbf{D}_{\infty h}(M)$ respectively; HOOD and HCCF have the same MS group but it is called $\mathbf{G}_2$ or $\mathbf{C}_{\infty v}(M)$ respectively. Figure 2 gives examples of how the MS group for four-atomic molecules with torsional tunnelling is named; the two character tables [4] are given in table 1. We will not consider $\mathbf{C}_{\infty v}(M)$ any further: all analysis relating to $\mathbf{D}_{\infty h}(M)$ will apply to $\mathbf{C}_{\infty v}(M)$ if the g, u labels are dropped.

In table 2, the transformation properties of the sequential bonding coordinates and some vibrational basis functions in $\mathbf{D}_{\infty h}(M)$ are given. The fact that each

Table 1. Character tables for $\mathbf{D}_{\infty h}(\mathrm{M})$, $\mathbf{C}_{\infty v}(\mathrm{M})$ and isomorphic groups.

$\mathbf{D}_{\infty h}(\mathrm{M})$	E	(14)(23)	E^*	(14)(23)*	$\mathbf{G}_4$	$\mathbf{C}_{2v}(\mathrm{M})$	$\mathbf{C}_{2h}(\mathrm{M})$
Σ_g^+	1	1	1	1	A_1	A_1	A_g
Σ_u^-	1	1	−1	−1	A_2	A_2	A_u
Σ_g^-	1	−1	−1	1	B_1	B_1	B_g
Σ_u^+	1	−1	1	−1	B_2	B_2	B_u

$\mathbf{C}_{\infty v}(\mathrm{M})$	E	E^*	$\mathbf{G}_2$	$\mathbf{C}_s(\mathrm{M})$
Σ^+	1	1	A'	A'
Σ^-	1	−1	A''	A''

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coordinate is mixed by elements of the MS group with at most only one other coordinate is a desirable feature resulting from vibrational coordinates well adapted to the underlying symmetry of $\hat{H}_{\mathrm{VR}}$ (see section 2.1). It facilitates the exploitation of symmetry during our procedure [24, 42, 43] of obtaining contracted basis functions of successively higher dimensionality through intermediate diagonalization. For example, in H_2O_2 a two-dimensional stretching Hamiltonian dependent only on $\{r_1, r_2\}$ can be blocked into Σ_g^+, Σ_u^+ by using a basis of simple symmetry combinations of product functions. Functions of (−) symmetry enter only through the torsion ϕ.

Before discussing rotational symmetry, we establish the notation $|J, K\rangle$ for a symmetric top function where K is the eigenvalue of $\hat{J}_z$ (in atomic units), and the value of the quantum number M is immaterial in the absence of external fields, and has been

Table 2. Transformation properties of sequential bonding coordinates and vibrational basis functions in $\mathbf{D}_{\infty h}(\mathrm{M})$.

E	(14)(23)	E^*	(14)(23)*	
r_1	r_2	r_1	r_2	$\Sigma_g^+ \oplus \Sigma_u^+$
r_2	r_1	r_2	r_1	
r_3	r_3	r_3	r_3	Σ_g^+
θ_1	θ_2	θ_1	θ_2	$\Sigma_g^+ \oplus \Sigma_u^+$
θ_2	θ_1	θ_2	θ_1	
ϕ	ϕ	$2\pi - \phi$	$2\pi - \phi$	Σ_u^-

$f(r_1)g(r_2) + f(r_2)g(r_1)$	Σ_g^+
$f(r_1)g(r_2) - f(r_2)g(r_1)$	Σ_u^+
$f(\theta_1)g(\theta_2) + f(\theta_2)g(\theta_1)$	Σ_g^+
$f(\theta_1)g(\theta_2) - f(\theta_2)g(\theta_1)$	Σ_u^+
$f(r_3)$	Σ_g^+
$\cos(m\phi)$	Σ_g^+
$\sin(m\phi)$	Σ_u^-
$\cos\left(\frac{(2m+1)}{2}\phi\right)$	Σ_u^-
$\sin\left(\frac{(2m+1)}{2}\phi\right)$	Σ_g^+

f and g are any functions and m is an integer.

Table 3. Equivalent rotations of $\mathbf{D}_{\infty h}(M)$ and transformation properties of $|J, K\rangle$ (and $\{\hat{J}_x, \hat{J}_y, \hat{J}_z\}$) in frames (a) and (b) (see figure 3).

Frame (a):

E	(14)(23)	E^*	(14)(23)*
R^0	$R^{\pi}_{\phi/2}$	$R^{\pi}_{\pi/2}$	$R^{\phi-\pi}_z$
$\|J, K\rangle$	$(-1)^J e^{-iK\phi}\|J, -K\rangle$	$(-1)^{J+K}\|J, -K\rangle$	$(-1)^K e^{iK\phi}\|J, K\rangle$

Frame (b):

E	(14)(23)	E^*	(14)(23)*	
R^0	R^{π}_0	R^{π}_0	R^0	
$\|J, K\rangle$	$(-1)^J\|J, -K\rangle$	$(-1)^J\|J, -K\rangle$	$\|J, K\rangle$	
$\hat{J}_x$	$\hat{J}_x$	$\hat{J}_x$	$\hat{J}_x$	Σ_g^+
$\hat{J}_y$	$-\hat{J}_y$	$-\hat{J}_y$	$\hat{J}_y$	Σ_g^-
$\hat{J}_z$	$-\hat{J}_z$	$-\hat{J}_z$	$\hat{J}_z$	Σ_g^-

Symmetrized functions:

$\|J, K\rangle + (-1)^J\|J, -K\rangle$	Σ_g^+
$\|J, K\rangle - (-1)^J\|J, -K\rangle$	Σ_g^-

dropped. In order to classify the rotational basis functions $|J, K\rangle$ and angular momentum operators $\{\hat{J}_x, \hat{J}_y, \hat{J}_z\}$ in the MS group, it is necessary first to determine the *equivalent rotation* [4] corresponding to each MS group element, i.e. the rotation of the nuclear-fixed frame that occurs as a consequence of a permutation or permutation-inversion. The equivalent rotations depend on how the nuclear-fixed frame has been defined. Following Bunker [4], we use R^0 to denote the null rotation, R^{π}_{α} to denote a rotation of the nuclear-fixed (x, y, z) axes through π radians about an axis in the xy plane making an angle α to the x axis, and R^{β}_z to denote a rotation of the (x, y, z) axes through an angle β about the z axis; the following formulae are then used to determine the effect of each MS group element on $|J, K\rangle$ and $\{\hat{J}_x, \hat{J}_y, \hat{J}_z\}$:

$$\begin{aligned} R^{\pi}_{\alpha}|J, K\rangle &= (-1)^J e^{-2iK\alpha}|J, K\rangle, \\ R^{\beta}_z|J, K\rangle &= e^{iK\beta}|J, K\rangle, \end{aligned} \tag{29}$$

$$\begin{aligned} R^{\pi}_{\alpha}\hat{J}_x(R^{\pi}_{\alpha})^{-1} &= \hat{J}_x\cos 2\alpha + \hat{J}_y\sin 2\alpha, \\ R^{\beta}_z\hat{J}_x(R^{\beta}_z)^{-1} &= \hat{J}_x\cos\beta + \hat{J}_y\sin\beta, \\ R^{\pi}_{\alpha}\hat{J}_y(R^{\pi}_{\alpha})^{-1} &= \hat{J}_x\sin 2\alpha - \hat{J}_y\cos 2\alpha, \\ R^{\beta}_z\hat{J}_y(R^{\beta}_z)^{-1} &= -\hat{J}_x\sin\beta + \hat{J}_y\cos\beta, \\ R^{\pi}_{\alpha}\hat{J}_z(R^{\pi}_{\alpha})^{-1} &= -\hat{J}_z, \\ R^{\beta}_z\hat{J}_z(R^{\beta}_z)^{-1} &= \hat{J}_z. \end{aligned} \tag{30}$$

In table 3 we give the equivalent rotations of $\mathbf{D}_{\infty h}(M)$ and, using (29), the transformation properties of $|J, K\rangle$ for the two choices of the nuclear-fixed frame shown

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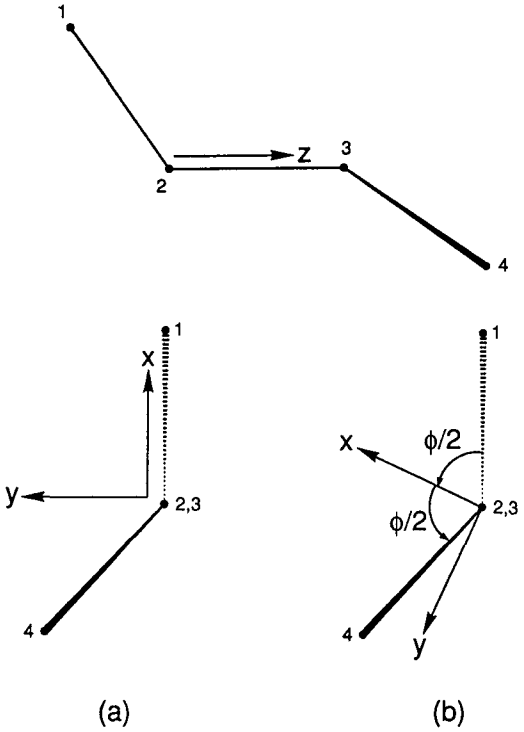


Figure 3. Two choices of the nuclear-fixed frame.

in figure 3. The first (a) is the frame used in [1] to derive $\hat{T}_{\mathrm{VR}}$ for a four-atomic molecule in sequential bonding coordinates, and the second (b) is a frame more adapted to the underlying symmetry of $\hat{H}_{\mathrm{VR}}$, which we have used (with the same vibrational coordinates) to derive our new $\hat{T}_{\mathrm{VR}}$ given in the Appendix. In frame (a), the $|J, K\rangle$ have characters that depend on the torsion angle ϕ, which means that to symmetrize the $|J, K\rangle$ would involve mixing in functions of ϕ, a vibrational coordinate. In frame (b) on the other hand, ϕ is not involved: symmetrized rotation functions are familiar Wang-like combinations [44], and $\{\hat{J}_x, \hat{J}_y, \hat{J}_z\}$ transform simply (using (30)). The new $\hat{T}_{\mathrm{VR}}$ is obviously preferable in practice, because it enables symmetry to be far more easily exploited in the parts of a variational calculation where vibration and rotation are coupled.

At first sight, frame (b) appears to be equivalent to that proposed for H_2O_2 in the internal axis method (IAM) of Hunt *et al.* [45], and by Bunker [46, 4] for dimethylacetylene (if the methyl groups are thought of as single atoms). It has been widely recognized [47] that if the nuclear-fixed axes are defined according to the IAM, then an extended MS (EMS) group is necessary to classify separately functions of the vibrational and rotational coordinates; this is because, in the IAM, the Euler angle γ and torsion angle ϕ are double-valued functions of the nuclear configuration, being defined respectively as $(\gamma_1 + \gamma_2)/2$ and $(\gamma_1 - \gamma_2)/2$, where γ_1 and γ_2 are Euler angles 'local' to each end of the molecule [46, 4]: the addition of 2π to γ_1 or γ_2 does not affect the nuclear configuration, but changes γ and ϕ (both of which have the range $(0, 2\pi)$) by π. The torsion–rotation coordinate system has the further curious property, when defined in this way, that if the molecule is held rigid and rotated about the

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(nuclear-fixed) z axis, ϕ undergoes a jump of π whenever γ_1 or γ_2 crosses the 0, 2π boundary. While retaining the useful consequences of symmetric orientation of the nuclear-fixed axes (see above), our definition of torsion–rotation coordinates (using frame (b)) is more straightforward than the IAM: we make no use of the concept of 'local' Euler angles; ϕ is defined very simply as in figure 1, and the value of γ follows from the location of the nuclear-fixed x axis halfway through the range of ϕ (see figure 3). Our coordinates are single-valued functions of the nuclear configuration, and so possess the useful property that vibrational and rotational basis functions *can* be classified separately in the ordinary MS group.

For A–B–A three-atomic molecules (i.e. those with MS group $\mathbf{D}_{\infty h}(\mathrm{M})$), there is a similar choice between having a nuclear-fixed axis along a bond, or along the bisector of the bond angle θ [1, 40, 48, 49]. Only the latter permits symmetrized $|J, K\rangle$ functions that do not depend on θ.

Previous accounts of the use of symmetry in variational rovibrational calculations have not made clear the inappropriateness of the point group, and have often lacked a clear explanation of the symmetry classification of basis functions. In a recent study of H_2O_2 [41], vibrational energy levels were labelled by $\mathbf{G}_4$-like symmetry labels without stating which symmetry group they referred to. If read as $\mathbf{G}_4$ labels as defined in table 2 and [4], they have been applied incorrectly.

Apart from the MS group, another true symmetry group of $\hat{H}_{VR}$ is **K** (spatial) [4], the group of all rotations of the molecule about all space-fixed axes through its centre of mass; its irreducible representations are labelled by the rotational quantum numbers J and M. Functions of vibrational coordinates and derivatives with respect to them commute with all the elements of **K** (spatial). The $\{\hat{J}_x, \hat{J}_y, \hat{J}_z\}$ are equivalent to infinitesimal rotations about nuclear-fixed axes, and since rotations about nuclear-fixed axes commute with rotations about space-fixed axes [11], the $\{\hat{J}_x, \hat{J}_y, \hat{J}_z\}$ too commute with all the elements of **K** (spatial). This is a proof of the remark in [1] that $\hat{T}_{VR}$ cannot contain the Euler angles at the end of the derivation: the Euler angles are not invariant to rotations about space-fixed axes.

4. The choice of basis functions

It is essential that the basis set $\{\Phi_i\}$ chosen for a variational calculation should be *completable*. We use this word to mean that the basis set becomes mathematically complete in its systematic extension towards infinite size. That is to say [50] that given ε and any continuous integrable function f, there exists a basis of size N such that

$$\left\langle f - \sum_{i=1}^{N} c_i\Phi_i \middle| f - \sum_{i=1}^{N} c_i\Phi_i \right\rangle < \varepsilon. \tag{31}$$

The choice of nuclear coordinate system can, however, impose important restraints on a basis set which may at first sight seem suitable. One kind of restraint arises from the choice of basis functions having the properties of continuity and single-valuedness possessed by 'physically admissable' [5] eigenfunctions of $\hat{H}_{VR}$. Another is connected with the problem of singularities. It is true of most nuclear coordinate systems that there are configurations of the nuclei for which one or more coordinate is undefined; or, equivalently, for which the Jacobian for the transformation from space-fixed cartesians is zero; or for which $\hat{H}_{VR}$ becomes infinite. Such singular regions always exist if Euler angles are used as rotational coordinates, because when all the nuclei are collinear, three Euler angles cannot be defined. Similar problems occur with the vibrational coordinates when, for instance, bond lengths are zero and

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bond angles become undefined; in our four-atomic example of sequential bonding coordinates, ϕ is undefined when either of θ_1 or θ_2 is zero. It is well known that the existence of singular regions in a rovibrational coordinate system causes singularities in $\hat{T}_{\mathrm{VR}}$, i.e. terms which become infinite in the singular regions. The basis set must be chosen to be completable while excluding functions that fail to cancel the singular terms in $\hat{T}_{\mathrm{VR}}$ when integrals across $\hat{H}_{\mathrm{VR}}$ are formed. This is discussed in more detail in section 4.2 below.

The general subject of singularities in rovibrational coordinate systems has been discussed by Sutcliffe [49, 51], who remarked that if two different forms of $\hat{H}_{\mathrm{VR}}$ are obtained in two correspondingly different coordinate systems by exact transformation from the same original form, then the two transformed operators need not have the same domain. We note that this statement remains true without preventing the two operators from having the same eigenvalues, because, in the integral representation of the Schrödinger equation, eigenfunctions are multiplied by the Jacobian, which differs in different coordinate systems; in particular, the regions in which it is zero differ.

Using our four-atomic system of sequential bonding coordinates and nuclear-fixed frame (b) (see figure 3) as an example, we consider in turn the two kinds of restraint on the basis set, and discuss how they affect the design of an efficient procedure to couple vibrational and rotational basis functions, keeping in mind that intricate restraints on the basis set often prove to be computationally highly inconvenient.

4.1. *Continuity and single-valuedness*

'Physically admissable' eigenfunctions of $\hat{H}_{\mathrm{VR}}$ are continuous and single-valued everywhere except in its singular regions [5]; basis functions can, therefore, be restricted to a completable set that shares this property. It is, however, surprisingly easy to overlook subtle discontinuities in apparently reasonable basis functions. These can occur at physically equivalent points in a subspace of the full rovibrational coordinate space, usually at the extremities of the range of one of the coordinates. A trivial example is a planar rotor described by a single angle $0 < \theta < 2\pi$, where the points $\theta = 0, 2\pi$ are equivalent, and any eigenfunction has the same value at the two points.

In our four-atomic example, there is a pair of equivalent points in the (ϕ, γ) space, where ϕ is the torsion angle and γ the Euler angle describing orientation about the nuclear-fixed z axis. Moving ϕ through the range $0 \rightarrow 2\pi$ is equivalent to adding π to γ, if all other coordinates are kept fixed in each case. Thus $(0, \gamma)$ and $(2\pi, \gamma \pm \pi)$ are equivalent points (where the $\pm$ allows γ to stay within its range of $(0, 2\pi)$). If $|J, K\rangle$ are used as rotational basis functions, and $e^{i\omega\phi}$ as torsional basis functions, then products $|J, K\rangle e^{i\omega\phi}$ should be made according to the rules

$$\left.\begin{aligned} K\,(\text{odd}) &\Rightarrow \omega = (2m + 1)/2 \\ K\,(\text{even}) &\Rightarrow \omega = m \end{aligned}\right\} \quad m = 0, 1, 2, \ldots \tag{32}$$

in order to be continuous.

It may also be tempting to apply this kind of argument to the angles θ_1 and θ_2; for example by noting that in (θ_1, ϕ) space, the points (θ_1, ϕ) and $(-\theta_1, \phi \pm \pi)$ are equivalent. This is erroneous, however, since one of these points must be outside the $(0, \pi)$ range of θ_1: the wavefunction is defined only within the range of the coordinates.

Arguments against certain basis functions can be made on the grounds that they should be single-valued throughout space. For example, note that the limiting

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procedure $\theta \to \pi$ (where θ means θ_1 or θ_2) leads to the same nuclear configuration irrespective of the value of ϕ. Consider a bending-torsion function $f(\theta)g(\phi)$ where $f(\theta)$ does not vanish when $\theta = \pi$. One can argue that since fg has different values in the limit $\theta \to \pi$ when the limit is taken at different values of ϕ, functions fg should be excluded from the basis. This argument is actually incorrect because the point in question is in a singular region of $\hat{H}_{VR}$ (see above). The value of the wavefunction in singular regions is irrelevant since, in the integral representation of the Schrödinger equation, it is removed by a zero Jacobian.

4.2. *Singularities in* $\hat{T}_{VR}$

One can distinguish two kinds of singular term $\hat{S}$ occurring in $\hat{T}_{VR}$. Some remain infinite when integrated, i.e. $\langle \Phi | \hat{S} | \Phi' \rangle$ is infinite where Φ, Φ' are some basis functions. Other have finite integrals, but basis functions Φ have infinite local energy with respect to them at singular points, i.e. the quotient $(\hat{S}\Phi)/\Phi$ is infinite. Of course if Ψ is an exact eigenfunction of the Hamiltonian $\hat{H}$, we know that $\langle \Psi | \hat{H} | \Psi \rangle$ is finite and that $(\hat{H}\Psi)/\Psi$ is equal to a constant energy at all values of the coordinates.

Both of these kinds of $\hat{S}$ are present in our new four-atomic $\hat{T}_{VR}$ (see Appendix), which has singular regions in vibrational space wherever the vibrational part of the Jacobian $r_1^2 r_2^2 r_3^2 \sin\theta_1 \sin\theta_2$ is zero. In previous work [43] using the purely vibrational part $\hat{T}_V$, attention was focused on those terms containing $(\cot\theta)\,\partial/\partial\theta$ and $\mathrm{cosec}^2\theta$ (where θ means θ_1 or θ_2). In the bending basis set $\{P_n^l(\cos\theta)$: $l = 0, 1, 2, \ldots; n = l, l+1, l+2, \ldots\}$ (associated Legendre functions), functions for which $l > 0$ have infinite local energy at $\theta = 0$ with respect to $(\cot\theta)\,\partial/\partial\theta$ through a $\mathrm{cosec}^2\theta$ term:

$$\cot\theta \frac{\partial}{\partial\theta} P_n^l = \left(l^2 \mathrm{cosec}^2\theta - \frac{\partial^2}{\partial\theta^2} - n(n+1) \right) P_n^l. \tag{33}$$

The integrals across $(\cot\theta)\,\partial/\partial\theta$ are, however, all finite, because of the $\sin\theta$ factor in the Jacobian. This is in contrast to the $\mathrm{cosec}^2\theta$ terms, for which the integrals $\{\langle P_n^0 | \mathrm{cosec}^2\theta | P_{n'}^0 \rangle\}$ are infinite. (Note that $P_n^l(\cos\theta)$ is equal to $\sin^l\theta$ multiplied by a polynomial in $\cos\theta$.) In [43], and earlier work on three-atomic molecules [49, 51], the two types of $\hat{S}$ were not distinguished, both types being considered in the basis set design; we have found, however, that in calculations using the purely vibrational part $\hat{T}_V$ of our four-atomic $\hat{T}_{VR}$, the basis need not be restricted to remove basis functions having infinite local energy, in particular, with respect to $(\cot\theta)\,\partial/\partial\theta$. This finding is not surprising: it was never proposed to omit any others of the many basis functions having infinite local energy with respect to many of the terms. In fact it has been shown frequently that so long as they do not have infinite integrals, basis functions having infinite local energy can be used successfully in variational calculations. This is the experience of electronic structure calculations using Gaussian basis functions, which have infinite local energy when the coordinates of any two particles are equal.

Some workers have found, in addition, that it is unnecessary to omit basis functions having infinite integrals, as long as the integrals are evaluated by quadrature, even if the singular points are near equilibrium (where one expects the eigenfunctions to have significantly non-zero amplitude) (see, for example, [33]). Without denying that such an approach has apparently been successful, we believe that it should be treated with extreme caution. Typically, there will be enough quadrature points n for the quadrature scheme to evaluate exactly all the integrals except the infinite ones; these are calculated instead to be some large number I. Then the only effect of changing n is to change the value of I: the greater n, the greater $|I|$. There may be no

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stability in the lowest m eigenvalues (i.e. those of interest) between the case of large n, in which the occurrence of very large numbers in a matrix diagonalization leads to loss of significant figures, and the case of small n, where finite components of infinite energy basis functions appear in the eigenvectors. Achieving a stable set of the lowest m eigenvalues at some intermediate value of n should be a prerequisite for believing that they correspond to true variational results.

Of course integrals involving harmonic oscillator and Morse stretching basis functions across many of the terms in $\hat{T}_{\mathrm{VR}}$ are strictly infinite due to the singularities at $r = 0$ (where r means r_1, r_2 or r_3); one obtains finite values for these integrals by a quadrature whose points in practice will all be far from $r = 0$. The justification is that all the integrals so calculated are numerically equivalent to those in an imaginary case in which the basis functions have been modified very slightly so as to vanish strongly near $r = 0$. The procedure is valid in this case only because the singularity is far from equilibrium, and all the eigenfunctions of interest have effectively zero amplitude there. The latter comment is also a justification for treating the basis functions as if they are orthogonal on the range $0 \leqslant r < \infty$.

In our four-atomic $\hat{T}_{\mathrm{VR}}$, ignoring the stretching singularities for the reasons just given, infinite integrals arise only from terms containing $\mathrm{cosec}^2\theta_1$ and $\mathrm{cosec}^2\theta_2$. These are

$$-\tfrac{1}{2}\mathrm{cosec}^2\theta_1\left(\frac{1}{\mu_1 r_1^2}+\frac{1}{\mu_3 r_3^2}-\frac{2\cos\theta_1}{M_2 r_1 r_3}\right)\left(\frac{\partial^2}{\partial\phi^2}-i\hat{J}_z\frac{\partial}{\partial\phi}-\frac{\hat{J}_z^2}{4}\right), \tag{34}$$

$$-\tfrac{1}{2}\mathrm{cosec}^2\theta_2\left(\frac{1}{\mu_2 r_2^2}+\frac{1}{\mu_3 r_3^2}-\frac{2\cos\theta_2}{M_3 r_2 r_3}\right)\left(\frac{\partial^2}{\partial\phi^2}+i\hat{J}_z\frac{\partial}{\partial\phi}-\frac{\hat{J}_z^2}{4}\right). \tag{35}$$

With a bending–torsion–rotation product basis set $\{P_{n_1}^{l_1}(\cos\theta_1)P_{n_2}^{l_2}(\cos\theta_2)\,\mathrm{e}^{\mathrm{i}\omega\phi}|J, K\rangle\}$, the infinite integrals $\{\langle P_n^0|\mathrm{cosec}^2\theta|P_{n'}^0\rangle\}$ can therefore be cancelled out by the following rules:

$$\begin{aligned}&\text{for } l_1 = 0, \quad -\omega^2+\omega K-\frac{K^2}{4}=0 \Rightarrow \omega=\frac{K}{2};\\ &\text{for } l_2 = 0, \quad -\omega^2-\omega K-\frac{K^2}{4}=0 \Rightarrow \omega=-\frac{K}{2}.\end{aligned} \tag{35}$$

Basis functions that violate the rules have infinite expectation values across $\hat{T}_{\mathrm{VR}}$. Note how rules (32) (derived from continuity considerations) and rules (35) are consistent. They are not, of course, a complete guide to the design of a bending–torsion–rotation basis, because $P_{n_1}^{l_1}$ and $P_{n_2}^{l_2}$ are non-orthogonal for $l_1 \neq l_2$: the basis set is completable while restricted to the use of only a few values of l_1 and l_2.

4.3. *The coupling of vibrational and rotational basis functions*

In variational rovibrational calculations on three-atomic molecules using valence coordinates, rules similar to (32) and (35) have been imposed in order to deal with singularities in $\hat{T}_{\mathrm{VR}}$ [48]. Within these restraints, one first obtains *vibrational* functions $\Phi_{\mathrm{vib}}^{(|K|)}$ optimized for each $|K|$, and then calculates the final rovibrational eigenvalues by using the basis set $\{\Phi_{\mathrm{vib}}^{(|K|)}|J, K\rangle, K = -J, \ldots, J\}$. The vibrational optimization to obtain $\Phi_{\mathrm{vib}}^{(|K|)}$ is done by diagonalizing H_{VR} in a basis consisting of vibrational functions multiplied by a single symmetrized $|J, K\rangle$ function [52].

This approach is unworkable for our new four-atomic $\hat{T}_{\mathrm{VR}}$, because a symmetrized basis for vibrational optimization in the manner just described could not satisfy our

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rules (32) and (35). In $\mathbf{D}_{\infty h}(M)$, the only symmetrized bending–torsion–rotation product functions allowed by the rules are

case (a): $l_1 \neq 0$ and $l_2 \neq 0$:

$$(P_{n_1}^{l_1}(\cos\theta_1)P_{n_2}^{l_2}(\cos\theta_2) \pm P_{n_2}^{l_2}(\cos\theta_1)P_{n_1}^{l_1}(\cos\theta_2))(e^{i\omega\phi} \pm e^{-i\omega\phi})(|J, K\rangle \pm |J, -K\rangle),$$

where K, ω obey rule (32);

case (b): $l_1 = 0$ or $l_2 = 0$:

$$(P_{n_1}^{l_1}(\cos\theta_1)P_{n_2}^{l_2}(\cos\theta_2) + P_{n_2}^{l_2}(\cos\theta_1)P_{n_1}^{l_1}(\cos\theta_2))(e^{iK\phi/2} + e^{-iK\phi/2})(|J, K\rangle + |J, -K\rangle),$$

$$+ (P_{n_1}^{l_1}(\cos\theta_1)P_{n_2}^{l_2}(\cos\theta_2) - P_{n_2}^{l_2}(\cos\theta_1)P_{n_1}^{l_1}(\cos\theta_2))(e^{iK\phi/2} - e^{-iK\phi/2})(|J, K\rangle - |J, -K\rangle). \quad (36)$$

(In case (a), the $\pm$ signs are to be taken as independent in order to generate all four symmetry types; in case (b), only one symmetry type is shown.) In view of this intricate coupling required of the rotational, torsional and bending basis functions, not to mention the fact that for linear and quasi-linear molecules the bending is more strongly coupled to rotation about the z axis than to the stretching motion, we suggest that the rotational, torsional and bending bases should be optimized together using a version of $\hat{T}_{VR}$ in which r_1, r_2, r_3 are fixed at their equilibrium values; and then products of optimized bending–torsion–rotation functions and three-dimensional stretching functions should form the final basis set.

5. Summary

In this paper we have reviewed the important criteria that affect the choice of that part of the molecular coordinate system describing nuclear motion, particularly with regard to the calculation of bound rovibrational states. We have expanded on a method [1] for deriving the nuclear kinetic energy operator, using as an example the derivation of an operator suitable for studying sequentially bonded four-atomic molecules. Following the approach to molecular symmetry expounded by Bunker [4], we have illustrated the use of the MS group in providing a rigorous but very simple symmetry classification of our four-atomic rovibrational coordinates, functions of them, and the kinetic energy operator. Finally, we have clarified some issues concerning boundary conditions and singularities in the design of a basis set for variational calculations, again using the sequentially bonded four-atomic molecule as an example. Variational calculations, using the kinetic energy operator presented in the Appendix, of the rovibrational spectra of C_2H_2, H_2O_2, H_2S_2, HCCF and other molecules are currently in progress.

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Appendix
A new $\hat{T}_{VR}$ operator for sequentially bonded four-atomic molecules

Terms in $\hat{T}_{VR}$ are obtained by multiplying each operator by $-\frac{1}{2}$ and the corresponding prefactor. We have taken $\hbar = 1$.

Nuclear masses are M_1, M_2, M_3, M_4. The reduced masses μ_1, μ_2, μ_3 are then defined as

$$\frac{1}{\mu_1} = \frac{1}{M_1} + \frac{1}{M_2}; \quad \frac{1}{\mu_2} = \frac{1}{M_3} + \frac{1}{M_4}; \quad \frac{1}{\mu_3} = \frac{1}{M_2} + \frac{1}{M_3}.$$

$$1 : \frac{2\cos\theta_1}{M_2 r_1 r_3} + \frac{2\cos\theta_2}{M_3 r_2 r_3}$$

$$\frac{\partial}{\partial r_1} : -\frac{2\cos\theta_1}{M_2 r_3}$$

$$\frac{\partial^2}{\partial r_1^2} : \frac{1}{\mu_1}$$

$$\frac{\partial^2}{\partial r_1 \partial r_2} : 0$$

$$\frac{\partial^2}{\partial r_1 \partial r_3} : \frac{2\cos\theta_1}{M_2}$$

$$\frac{\partial^2}{\partial r_1 \partial \theta_1} : -\frac{2\sin\theta_1}{M_2 r_3}$$

$$\frac{\partial^2}{\partial r_1 \partial \theta_2} : \frac{2\cos\phi\sin\theta_1}{M_2 r_3}$$

$$\frac{\partial^2}{\partial r_1 \partial \phi} : -\frac{2\sin\phi\sin\theta_1\cot\theta_2}{M_2 r_3}$$

$$\frac{\partial}{\partial r_2} : -\frac{2\cos\theta_2}{M_3 r_3}$$

$$\frac{\partial^2}{\partial r_2^2} : \frac{1}{\mu_2}$$

$$\frac{\partial^2}{\partial r_2 \partial r_3} : \frac{2\cos\theta_2}{M_3}$$

$$\frac{\partial^2}{\partial r_2 \partial \theta_1} : \frac{2\cos\phi\sin\theta_2}{M_3 r_3}$$

$$\frac{\partial^2}{\partial r_2 \partial \theta_2} : -\frac{2\sin\theta_2}{M_3 r_3}$$

$$\frac{\partial^2}{\partial r_2 \partial \phi} : -\frac{2\sin\phi\cot\theta_1\sin\theta_2}{M_3 r_3}$$

$$\frac{\partial}{\partial r_3} : -\frac{2\cos\theta_1}{M_2 r_1} - \frac{2\cos\theta_2}{M_3 r_2}$$

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$$\frac{\partial^2}{\partial r_3^2} : \frac{1}{\mu_3}$$

$$\frac{\partial^2}{\partial r_3 \partial \theta_1} : -\frac{2\sin\theta_1}{M_2 r_1}$$

$$\frac{\partial^2}{\partial r_3 \partial \theta_2} : -\frac{2\sin\theta_2}{M_3 r_2}$$

$$\frac{\partial^2}{\partial r_3 \partial \phi} : 0$$

$$\frac{\partial}{\partial \theta_1} : -\frac{2\cos\phi\sin\theta_2}{M_3 r_2 r_3} + \cot\theta_1\left(\frac{1}{\mu_1 r_1^2} + \frac{1}{\mu_3 r_3^2}\right) + \frac{4\sin\theta_1}{M_2 r_1 r_3} - \frac{2\operatorname{cosec}\theta_1}{M_2 r_1 r_3}$$

$$\frac{\partial^2}{\partial \theta_1^2} : -\frac{2\cos\theta_1}{M_2 r_1 r_3} + \frac{1}{\mu_1 r_1^2} + \frac{1}{\mu_3 r_3^2}$$

$$\frac{\partial^2}{\partial \theta_1 \partial \theta_2} : 2\cos\phi\left(\frac{\cos\theta_1}{M_2 r_1 r_3} + \frac{\cos\theta_2}{M_3 r_2 r_3} - \frac{1}{\mu_3 r_3^2}\right)$$

$$\frac{\partial^2}{\partial \theta_1 \partial \phi} : 2\sin\phi\left(-\frac{\cos\theta_1\cot\theta_2}{M_2 r_1 r_3} + \frac{\cot\theta_2}{\mu_3 r_3^2} - \frac{\operatorname{cosec}\theta_2}{M_3 r_2 r_3}\right)$$

$$\frac{\partial}{\partial \theta_2} : -\frac{2\cos\phi\sin\theta_1}{M_2 r_1 r_3} + \cot\theta_2\left(\frac{1}{\mu_2 r_2^2} + \frac{1}{\mu_3 r_3^2}\right) + \frac{4\sin\theta_2}{M_3 r_2 r_3} - \frac{2\operatorname{cosec}\theta_2}{M_3 r_2 r_3}$$

$$\frac{\partial^2}{\partial \theta_2^2} : -\frac{2\cos\theta_2}{M_3 r_2 r_3} + \frac{1}{\mu_2 r_2^2} + \frac{1}{\mu_3 r_3^2}$$

$$\frac{\partial^2}{\partial \theta_2 \partial \phi} : 2\sin\phi\left(-\frac{\cot\theta_1\cos\theta_2}{M_3 r_2 r_3} + \frac{\cot\theta_1}{\mu_3 r_3^2} - \frac{\operatorname{cosec}\theta_1}{M_2 r_1 r_3}\right)$$

$$\frac{\partial}{\partial \phi} : 2\sin\phi\left(\frac{\cot\theta_1\sin\theta_2}{M_3 r_2 r_3} + \frac{\sin\theta_1\cot\theta_2}{M_2 r_1 r_3}\right)$$

$$\frac{\partial^2}{\partial \phi^2} : \frac{2\cos\phi\cot\theta_1\cot\theta_2}{\mu_3 r_3^2} - \frac{2\cos\phi\cot\theta_1\operatorname{cosec}\theta_2}{M_3 r_2 r_3} - \frac{2\cos\phi\operatorname{cosec}\theta_1\cot\theta_2}{M_2 r_1 r_3}$$

$$- \frac{2\cot\theta_1\operatorname{cosec}\theta_1}{M_2 r_1 r_3} - \frac{2\cot\theta_2\operatorname{cosec}\theta_2}{M_3 r_2 r_3} - \frac{2}{\mu_3 r_3^2} + \operatorname{cosec}^2\theta_2\left(\frac{1}{\mu_2 r_2^2} + \frac{1}{\mu_3 r_3^2}\right)$$

$$+ \operatorname{cosec}^2\theta_1\left(\frac{1}{\mu_1 r_1^2} + \frac{1}{\mu_3 r_3^2}\right)$$

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$$\frac{\partial}{\partial r_1} \mathrm{i}\hat{J}_x : \frac{2 \sin(\phi/2) \sin\theta_1}{M_2 r_3}$$

$$\frac{\partial}{\partial r_1} \mathrm{i}\hat{J}_y : \frac{2 \cos(\phi/2) \sin\theta_1}{M_2 r_3}$$

$$\frac{\partial}{\partial r_1} \mathrm{i}\hat{J}_z : -\frac{\sin\phi \sin\theta_1 \cot\theta_2}{M_2 r_3}$$

$$\frac{\partial}{\partial r_2} \mathrm{i}\hat{J}_x : \frac{2 \sin(\phi/2) \sin\theta_2}{M_3 r_3}$$

$$\frac{\partial}{\partial r_2} \mathrm{i}\hat{J}_y : -\frac{2 \cos(\phi/2) \sin\theta_2}{M_3 r_3}$$

$$\frac{\partial}{\partial r_2} \mathrm{i}\hat{J}_z : \frac{\sin\phi \cot\theta_1 \sin\theta_2}{M_3 r_3}$$

$$\frac{\partial}{\partial r_3} \mathrm{i}\hat{J}_x : 0$$

$$\frac{\partial}{\partial r_3} \mathrm{i}\hat{J}_y : 0$$

$$\frac{\partial}{\partial r_3} \mathrm{i}\hat{J}_z : 0$$

$$\frac{\partial}{\partial \theta_1} \mathrm{i}\hat{J}_x : 2 \sin(\phi/2) \left(\frac{\cos\theta_1}{M_2 r_1 r_3} - \frac{1}{\mu_3 r_3^2} \right)$$

$$\frac{\partial}{\partial \theta_1} \mathrm{i}\hat{J}_y : 2 \cos(\phi/2) \left(\frac{\cos\theta_1}{M_2 r_1 r_3} - \frac{1}{\mu_3 r_3^2} \right)$$

$$\frac{\partial}{\partial \theta_1} \mathrm{i}\hat{J}_z : \sin\phi \left(-\frac{\cos\theta_1 \cot\theta_2}{M_2 r_1 r_3} + \frac{\cot\theta_2}{\mu_3 r_3^2} - \frac{\mathrm{cosec}\,\theta_2}{M_3 r_2 r_3} \right)$$

$$\frac{\partial}{\partial \theta_2} \mathrm{i}\hat{J}_x : 2 \sin(\phi/2) \left(\frac{\cos\theta_2}{M_3 r_2 r_3} - \frac{1}{\mu_3 r_3^2} \right)$$

$$\frac{\partial}{\partial \theta_2} \mathrm{i}\hat{J}_y : -2 \cos(\phi/2) \left(\frac{\cos\theta_2}{M_3 r_2 r_3} - \frac{1}{\mu_3 r_3^2} \right)$$

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$$\frac{\partial}{\partial\theta_2}\,\mathrm{i}\hat{J}_z : \sin\phi\left(\frac{\cot\theta_1\cos\theta_2}{M_3r_2r_3} - \frac{\cot\theta_1}{\mu_3r_3^2} + \frac{\operatorname{cosec}\theta_1}{M_2r_1r_3}\right)$$

$$\frac{\partial}{\partial\phi}\,\mathrm{i}\hat{J}_x : 2\cos(\phi/2)\left(-\frac{\cot\theta_1}{\mu_3r_3^2} - \frac{\cot\theta_2}{\mu_3r_3^2} + \frac{\operatorname{cosec}\theta_2}{M_3r_2r_3} + \frac{\operatorname{cosec}\theta_1}{M_2r_1r_3}\right)$$

$$\frac{\partial}{\partial\phi}\,\mathrm{i}\hat{J}_y : 2\sin(\phi/2)\left(\frac{\cot\theta_1}{\mu_3r_3^2} - \frac{\cot\theta_2}{\mu_3r_3^2} + \frac{\operatorname{cosec}\theta_2}{M_3r_2r_3} - \frac{\operatorname{cosec}\theta_1}{M_2r_1r_3}\right)$$

$$\frac{\partial}{\partial\phi}\,\mathrm{i}\hat{J}_z : \frac{2\cot\theta_1\operatorname{cosec}\theta_1}{M_2r_1r_3} - \frac{2\cot\theta_2\operatorname{cosec}\theta_2}{M_3r_2r_3} + \operatorname{cosec}^2\theta_2\left(\frac{1}{\mu_2r_2^2} + \frac{1}{\mu_3r_3^2}\right)$$

$$- \operatorname{cosec}^2\theta_1\left(\frac{1}{\mu_1r_1^2} + \frac{1}{\mu_3r_3^2}\right)$$

$$\mathrm{i}\hat{J}_x : \frac{\sin(\phi/2)}{2}\left(-\frac{\cot\theta_1}{\mu_3r_3^2} - \frac{\cot\theta_2}{\mu_3r_3^2} - \frac{4\sin\theta_1}{M_2r_1r_3} - \frac{4\sin\theta_2}{M_3r_2r_3} + \frac{\operatorname{cosec}\theta_2}{M_3r_2r_3} + \frac{\operatorname{cosec}\theta_1}{M_2r_1r_3}\right)$$

$$\mathrm{i}\hat{J}_y : \frac{\cos(\phi/2)}{2}\left(-\frac{\cot\theta_1}{\mu_3r_3^2} + \frac{\cot\theta_2}{\mu_3r_3^2} - \frac{4\sin\theta_1}{M_2r_1r_3} + \frac{4\sin\theta_2}{M_3r_2r_3} - \frac{\operatorname{cosec}\theta_2}{M_3r_2r_3} + \frac{\operatorname{cosec}\theta_1}{M_2r_1r_3}\right)$$

$$\mathrm{i}\hat{J}_z : \sin\phi\left(-\frac{\cot\theta_1\sin\theta_2}{M_3r_2r_3} + \frac{\sin\theta_1\cot\theta_2}{M_2r_1r_3}\right)$$

$$\hat{J}_x^2 : -\frac{1}{\mu_3r_3^2}$$

$$\hat{J}_y^2 : -\frac{1}{\mu_3r_3^2}$$

$$\hat{J}_z^2 : \frac{\cos\phi\cot\theta_1\cot\theta_2}{2\mu_3r_3^2} - \frac{\cos\phi\cot\theta_1\operatorname{cosec}\theta_2}{2M_3r_2r_3} - \frac{\cos\phi\operatorname{cosec}\theta_1\cot\theta_2}{2M_2r_1r_3}$$

$$+ \frac{\cot\theta_1\operatorname{cosec}\theta_1}{2M_2r_1r_3} + \frac{\cot\theta_2\operatorname{cosec}\theta_2}{2M_3r_2r_3} + \frac{1}{2\mu_3r_3^2} - \frac{\operatorname{cosec}^2\theta_2}{4}\left(\frac{1}{\mu_2r_2^2} + \frac{1}{\mu_3r_3^2}\right)$$

$$- \frac{\operatorname{cosec}^2\theta_1}{4}\left(\frac{1}{\mu_1r_1^2} + \frac{1}{\mu_3r_3^2}\right)$$

$$[\hat{J}_x, \hat{J}_y]_+ : 0$$

$$[\hat{J}_x, \hat{J}_z]_+ : \frac{\cos(\phi/2)}{2}\left(-\frac{\cot\theta_1}{\mu_3r_3^2} + \frac{\cot\theta_2}{\mu_3r_3^2} - \frac{\operatorname{cosec}\theta_2}{M_3r_2r_3} + \frac{\operatorname{cosec}\theta_1}{M_2r_1r_3}\right)$$

$$[\hat{J}_y, \hat{J}_z]_+ : \frac{\sin(\phi/2)}{2}\left(\frac{\cot\theta_1}{\mu_3r_3^2} + \frac{\cot\theta_2}{\mu_3r_3^2} - \frac{\operatorname{cosec}\theta_2}{M_3r_2r_3} - \frac{\operatorname{cosec}\theta_1}{M_2r_1r_3}\right)$$

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